화학을위한수학 (2023년 1학기) **1차 시험 해답**

(총 6 문제, 100점)

1. **(총 20점, 각 5점)**

(a)

$$\begin{aligned} z &= 4e^{3i} + 5e^{2i} \\ &= 4(\cos 3 + i\sin 3) + 5(\cos 2 + i\sin 2) \\ &= (4\cos 3 + 5\cos 2) + i(4\sin 3 + 5\sin 2) \end{aligned}$$

$\therefore \mathrm{Re}(z) = 4\cos 3 + 5\cos 2, \qquad \mathrm{Im}(z) = 4\sin 3 + 5\sin 2$ 이다.

(b)

$$\begin{aligned} z^* &= \left(\frac{3+2i}{4+5i}\right)^2 = \left(\frac{3+2i}{4+5i} \times \frac{4-5i}{4-5i}\right)^2 \\ &= \left(\frac{12-15i+8i+10}{16+25}\right)^2 = \frac{(22-7i)^2}{41^2} \\ &= \frac{484-308i-49}{1681} = \frac{435}{1681} - i\frac{308}{1681} \end{aligned}$$

그러므로 $z = \frac{435}{1681} + i\frac{308}{1681}$.

$\therefore \mathrm{Re}(z) = \frac{435}{1681}, \qquad \mathrm{Im}(z) = \frac{308}{1681}$ 이다.

(c)

$$\begin{aligned} z = (\sqrt{3}-i)^{50} &= \left[2\left(\frac{\sqrt{3}}{2} - \frac{1}{2}\right)\right]^{50} \\ &= \left[2\Big(\cos\Big(-\frac{\pi}{6}\Big) + i\sin\Big(-\frac{\pi}{6}\Big)\Big)\right]^{50} \\ &= 2^{50}\left[\cos\Big(-50\times\frac{\pi}{6}\Big) + i\sin\Big(-50\times\frac{\pi}{6}\Big)\right] \text{ (De Moivre 공식에 의해)} \\ &= 2^{50}\left[\cos\Big(-\frac{25\pi}{3}\Big) + i\sin\Big(-\frac{25\pi}{3}\Big)\right] \\ &= 2^{50}\left[\cos\Big(-8\pi-\frac{\pi}{3}\Big) + i\sin\Big(-8\pi-\frac{\pi}{3}\Big)\right] \\ &= 2^{50}\left[\cos\Big(-\frac{\pi}{3}\Big) + i\sin\Big(-\frac{\pi}{3}\Big)\right] \\ &= 2^{50}\left(\frac{1}{2} - i\frac{\sqrt{3}}{2}\right) \end{aligned}$$

$\therefore \mathrm{Re}(z) = 2^{49}, \qquad \mathrm{Im}(z) = -2^{49}\sqrt{3}.$

(d) $z = a + bi$ (단, a와 b는 실수) 라면,

$$\begin{aligned} \mathrm{Re}(iz) + i\mathrm{Im}(z^*(2+i)) &= \mathrm{Re}(-b+ai) + \mathrm{Im}\big[(a-bi)(2+i)\big] \\ &= -b + i\mathrm{Im}(2a + ai - 2bi + b) \\ &= -b + (a-2b)i = -2 \qquad (1) \end{aligned}$$

복소수 상등정리를 이용하면 (1) 식으로부터 $-b = -2$, 그리고 $a - 2b = 0$. $\Rightarrow$ $b = 2$ 이고, $a = 4$ 이므로 $z = 4 + 2i$. $\mathrm{Re}(z) = 4$, $\mathrm{Im(z)} = 2$

2. **(총 10점, 각 5점)**

(a) $(r, \theta, \phi) = (2, -\pi/4, 3\pi/4)$

$$x = r\sin\theta\cos\phi = 2 \times \sin\left(-\frac{\pi}{4}\right) \times \cos\left(\frac{3\pi}{4}\right) = 2 \times \left(-\frac{\sqrt{2}}{2}\right) \times \left(-\frac{\sqrt{2}}{2}\right) = 1$$

$$y = r\sin\theta\sin\phi = 2 \times \sin\left(-\frac{\pi}{4}\right) \times \sin\left(\frac{3\pi}{4}\right) = 2 \times \left(-\frac{\sqrt{2}}{2}\right) \times \left(\frac{\sqrt{2}}{2}\right) = -1$$

$$y = r\cos\theta = 2 \times \cos\left(-\frac{\pi}{4}\right) = 2 \times \frac{\sqrt{2}}{2} = \sqrt{2}$$

(b) $(x, y, z) = (-2, 1, -3)$

$$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{(-2)^2 + 1^2 + (-3)^2} = \sqrt{14}$$

$$\theta = \cos^{-1}\left(\frac{z}{r}\right) = \cos^{-1}\left(-\frac{3}{\sqrt{14}}\right)$$

$$\phi = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(-\frac{1}{2}\right)$$

3. **(총 20점, 각 5점)**

(a) 문제의 (1) 식을 다시 쓰면

$$\psi_{321}(r, \theta, \phi) = \frac{1}{81\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{3/2}\left(\frac{r}{a_0}\right)^2 e^{-r/(3a_0)}\sin\theta\cos\theta e^{i\phi} \tag{2}$$

$$= \frac{1}{81\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{3/2}\left(\frac{r}{a_0}\right)^2 e^{-r/(3a_0)}\sin\theta\cos\theta(\cos\phi + i\sin\phi) \tag{3}$$

그러므로

$$\mathrm{Re}(\psi_{321}) = \frac{1}{81\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{3/2}\left(\frac{r}{a_0}\right)^2 e^{-r/(3a_0)}\sin\theta\cos\theta\cos\phi,$$

$$\mathrm{Im}(\psi_{321}) = \frac{1}{81\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{3/2}\left(\frac{r}{a_0}\right)^2 e^{-r/(3a_0)}\sin\theta\cos\theta\sin\phi$$

(b) (3) 식으로부터

$$\psi^*_{321}(r, \theta, \phi) = \frac{1}{81\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{3/2}\left(\frac{r}{a_0}\right)^2 e^{-r/(3a_0)}\sin\theta\cos\theta(\cos\phi - i\sin\phi)$$

$$= \frac{1}{81\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{3/2}\left(\frac{r}{a_0}\right)^2 e^{-r/(3a_0)}\sin\theta\cos\theta\big[\cos(-\phi) + i\sin(-\phi)\big]$$

$$= \frac{1}{81\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{3/2}\left(\frac{r}{a_0}\right)^2 e^{-r/(3a_0)}\sin\theta\cos\theta e^{-i\phi} \tag{4}$$

(c) (2)와 (4) 식으로부터

$$|\psi_{321}(r, \theta, \phi)|^2 = \psi_{321}\psi^*_{321} = \frac{1}{81^2\pi}\left(\frac{1}{a_0}\right)^3\left(\frac{r}{a_0}\right)^4 e^{-2r/(3a_0)}\sin^2\theta\cos^2\theta$$

(d) (a)의 (3) 식으로부터

$$\begin{aligned}\psi_{321}(r,\theta,\phi) &= \frac{1}{81\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{3/2}\left(\frac{r}{a_0}\right)^{2} e^{-r/(3a_0)}\sin\theta\cos\theta(\cos\phi + i\sin\phi) \\ &= \frac{1}{81\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{5/2} e^{-r/(3a_0)} r\cos\theta\big(r\sin\theta\cos\phi + ir\sin\theta\sin\phi\big)\end{aligned} \tag{5}$$

3차원 직교 좌표계와 구면 극 좌표계의 관계식으로부터

$$\begin{aligned}x &= r\sin\theta\cos\phi \\ y &= r\sin\theta\sin\phi \\ z &= r\cos\theta \\ r &= \sqrt{x^2+y^2+z^2}\end{aligned}$$

이므로, 관계식을 위의 (5) 식에 대입하면

$$\psi_{321}(x,y,z) = \frac{1}{81\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{5/2} e^{-\sqrt{x^2+y^2+z^2}/(3a_0)} z\big(x+iy\big)$$

4. **(총 10점, 각 5점)**

(a)

$$\begin{aligned}f_x(x,y) = \left(\frac{\partial f}{\partial x}\right)_y &= ay\ln(y) \\ f_y(x,y) = \left(\frac{\partial f}{\partial y}\right)_x &= ax\ln(y) + ax = ax[\ln(y)+1] \\ f_{xx}(x,y) = \left(\frac{\partial^2 f}{\partial x^2}\right)_y &= 0 \\ f_{yy}(x,y) = \left(\frac{\partial^2 f}{\partial y^2}\right)_x &= \frac{ax}{y} \\ f_{xy}(x,y) = \left(\frac{\partial^2 f}{\partial y\partial x}\right) &= a\ln(y) + a = a[\ln(y)+1]\end{aligned}$$

(b)

$$\begin{aligned}f_x(x,y) = \left(\frac{\partial f}{\partial x}\right)_y &= -ye^{-y}\sin(xy) \\ f_y(x,y) = \left(\frac{\partial f}{\partial y}\right)_x &= -e^{-y}\cos(xy) - xe^{-y}\sin(xy) \\ &= -e^{-y}[\cos(xy) + x\sin(xy)] \\ f_{xx}(x,y) = \left(\frac{\partial^2 f}{\partial x^2}\right)_y &= -y^2e^{-y}\cos(xy) \\ f_{yy}(x,y) = \left(\frac{\partial^2 f}{\partial y^2}\right)_x &= e^{-y}\cos(xy) + 2xe^{-y}\sin(xy) - x^2e^{-y}\cos(xy) \\ &= e^{-y}\big[(1-x^2)\cos(xy) + 2x\sin(xy)\big] \\ f_{xy}(x,y) = \left(\frac{\partial^2 f}{\partial y\partial x}\right) &= -e^{-y}\sin(xy) + ye^{-y}\sin(xy) - xye^{-y}\cos(xy) \\ &= e^{-y}\big[(y-1)\sin(xy) - xy\cos(xy)\big]\end{aligned}$$

5. **(총 20점, 각 5점)**

(a) $\sin(xy)dx + \sin(xy)dy$

$$M(x,y) = \sin(xy), \qquad N(x,y) = \sin(xy)$$
$$M_y(x,y) = \left(\frac{\partial M}{\partial y}\right)_x = x\cos(xy)$$
$$N_x(x,y) = \left(\frac{\partial N}{\partial x}\right)_y = y\cos(xy)$$
$$M_y(x,y) \neq N_x(x,y)$$

불완전미분

(b) $ye^{axy}dx + xe^{axy}dy$

$$M(x,y) = ye^{axy}, \qquad N(x,y) = xe^{axy}$$
$$M_y(x,y) = \left(\frac{\partial M}{\partial y}\right)_x = e^{axy} + axye^{axy}$$
$$N_y(x,y) = \left(\frac{\partial N}{\partial x}\right)_y = e^{axy} + axye^{axy}$$
$$M_y(x,y) = N_x(x,y)$$

완전미분, $f(x,y) = \frac{1}{a}e^{axy} + C$

(c) $xy^2dx + x^2ydy$

$$M(x,y) = xy^2, \qquad N(x,y) = x^2y$$
$$M_y(x,y) = \left(\frac{\partial M}{\partial y}\right)_x = 2xy$$
$$N_y(x,y) = \left(\frac{\partial N}{\partial x}\right)_y = 2xy$$
$$M_y(x,y) = N_x(x,y)$$

완전미분, $f(x,y) = \frac{x^2y^2}{2} + C$

(d) $C_V(T)dT + \frac{RT}{V}dV$

$$M(T,V) = C_V(T), \qquad N(x,y) = \frac{RT}{V}$$
$$M_V(T,V) = \left(\frac{\partial M}{\partial V}\right)_T = 0$$
$$N_T(T,V) = \left(\frac{\partial N}{\partial T}\right)_V = \frac{R}{V}$$
$$M_V(T,V) \neq N_T(T,V)$$

불완전 미분.

6. **(총 20점, 각 10점)**

(a) $f(x,y) = 2x - x^2 + 2y^2 - y^4$ 가 (x^*, y^*) 에서 극값을 가지므로

$$f_x(x^*,y^*) = 0 \Rightarrow 2 - 2x^* = 0 \Rightarrow x^* = 1$$
$$f_y(x^*,y^*) = 0 \Rightarrow 4y^* - 4(y^*)^3 = 0 \Rightarrow 4y^*(y^*+1)(y^*-1) = 0 \Rightarrow y^* = 0, \pm 1$$

그러므로 **임계점은 (1,0), (1, -1), (1, 1)** 이다. Hessian 검사를 이용하면

$$\Delta_1 = f_{xx}(x^*, y^*) \quad \text{그리고} \quad \Delta_2 = f_{xx}(x^*, y^*) f_{yy}(x^*, y^*) - f_{xy}(x^*, y^*)^2.$$

주어진 함수 $f(x, y) = 2x - x^2 + 2y^2 - y^4$ 에 대하여

$$\begin{aligned} f_{xx}(x, y) &= -2 \\ f_{yy}(x, y) &= 4 - 12y^2 \\ f_{xy}(x, y) &= 0 \end{aligned}$$

i. $(1, 0)$ 에서

$$\begin{aligned} f_{xx}(1, 0) &= -2 \\ f_{yy}(1, 0) &= 4 \\ f_{xy}(1, 0) &= 0 \end{aligned}$$

그러므로

$$\Delta_1 = -2 < 0 \quad \text{그리고} \quad \Delta_2 = (-2) \times (4) - (0)^2 = -8 < 0$$

$\Delta_2 < 0$ 이므로 $f(x, y)$ 는 $(1, 0)$ 에서 안장점을 갖는다.

ii. $(1, -1)$ 에서

$$\begin{aligned} f_{xx}(1, -1) &= -2 \\ f_{yy}(1, -1) &= -8 \\ f_{xy}(1, -1) &= 0 \end{aligned}$$

그러므로

$$\Delta_1 = -2 < 0 \quad \text{그리고} \quad \Delta_2 = (-2) \times (-8) - (0)^2 = 16 > 0$$

$\Delta_1 < 0$ 이고, $\Delta_2 > 0$ 이므로 $f(x, y)$ 는 $(1, -1)$ 에서 극댓값을 갖는다.

iii. $(1, 1)$ 에서

$$\begin{aligned} f_{xx}(1, 1) &= -2 \\ f_{yy}(1, 1) &= -8 \\ f_{xy}(1, 1) &= 0 \end{aligned}$$

그러므로

$$\Delta_1 = -2 < 0 \quad \text{그리고} \quad \Delta_2 = (-2) \times (-8) - (0)^2 = 16 > 0$$

$\Delta_1 < 0$ 이고, $\Delta_2 > 0$ 이므로 $f(x, y)$ 는 $(1, 1)$ 에서 극댓값을 갖는다.

(b) $f(x, y) = e^{2x - 4y - x^2 - y^2}$ 가 (x^*, y^*) 에서 극값을 가지므로

$$\begin{aligned} f_x(x^*, y^*) = 0 &\Rightarrow (2 - 2x^*) e^{2x^* - 4y^* - (x^*)^2 - (y^*)^2} = 0 \Rightarrow x^* = 1 \\ f_y(x^*, y^*) = 0 &\Rightarrow (-4 - 2y^*) e^{2x^* - 4y^* - (x^*)^2 - (y^*)^2} = 0 \Rightarrow y^* = -2 \end{aligned}$$

그러므로 **임계점은 (1,-2)** 이다. Hessian 검사를 이용하면

$$\Delta_1 = f_{xx}(x^*, y^*) \quad \text{그리고} \quad \Delta_2 = f_{xx}(x^*, y^*) f_{yy}(x^*, y^*) - f_{xy}(x^*, y^*)^2.$$

주어진 함수 $f(x, y) = e^{2x-4y-x^2-y^2}$ 에 대하여

$$\begin{aligned} f_{xx}(x, y) &= -2e^{2x-4y-x^2-y^2} + (2-2x)^2 e^{2x-4y-x^2-y^2} \\ &= (2 - 8x + 4x^2) e^{2x-4y-x^2-y^2} \\ f_{yy}(x, y) &= -2e^{2x-4y-x^2-y^2} + (-4-2y)^2 e^{2x-4y-x^2-y^2} \\ &= (14 + 16y + 4y^2) e^{2x-4y-x^2-y^2} \\ f_{xy}(x, y) &= (2-2x)(-4-2y) e^{2x-4y-x^2-y^2} \end{aligned}$$

임계점 $(1, -2)$ 에서

$$\begin{aligned} f_{xx}(1, -2) &= -2e^5 \\ f_{yy}(1, -2) &= -2e^5 \\ f_{xy}(1, -2) &= 0 \end{aligned}$$

그러므로

$$\Delta_1 = -2e^5 < 0 \quad \text{그리고} \quad \Delta_2 = (-2e^5) \times (-2e^5) - (0)^2 = 4e^{10} > 0$$

$\Delta_1 < 0$ 이고, $\Delta_2 > 0$ 이므로 $f(x, y)$ 는 $(1, -2)$ 에서 극댓값을 갖는다.